PRAYER

Dear Lord and Father of all, Thank you for today. Thank you for ways in which you provide for us all. For Your protection and love we thank you. Help us to focus our hearts and minds now on what we are about to learn. Inspire us by Your Holy Spirit as we listen and write. Guide us by your eternal light as we discover more about the world around us.

We ask all this in the name of Jesus.

Amen.

MATATAG CURRICULUM

TECHNOLOGY AND LIVELIHOOD EDUCATION

QUARTER 4

Short Review

Instructions:

Read each statement carefully. Write True if the statement is correct and False if it is incorrect.

1

The Philippine Electrical Code (PEC) only provides guidelines for electrical wiring and does not cover safety measures for grounding and protection.

FALSE

Short Review

2

Instructions:

Read each statement carefully. Write True if the statement is correct and False if it is incorrect.

RA 7920, also known as the New Electrical Engineering Law, mandates the use of the Philippine Electrical Code (PEC).

TRUE

Short Review

3

Instructions:

Read each statement carefully. Write True if the statement is correct and False if it is incorrect.

The Bureau of Fire Protection (BFP) has legal authority over the design of electrical systems as part of fire safety regulations.

FALSE

Short Review

4

Instructions:

Read each statement carefully. Write True if the statement is correct and False if it is incorrect.

A certificate of final electrical inspection (CFEI) is issued only after an inspection of electrical installations.

TRUE

Short Review

5

Instructions:

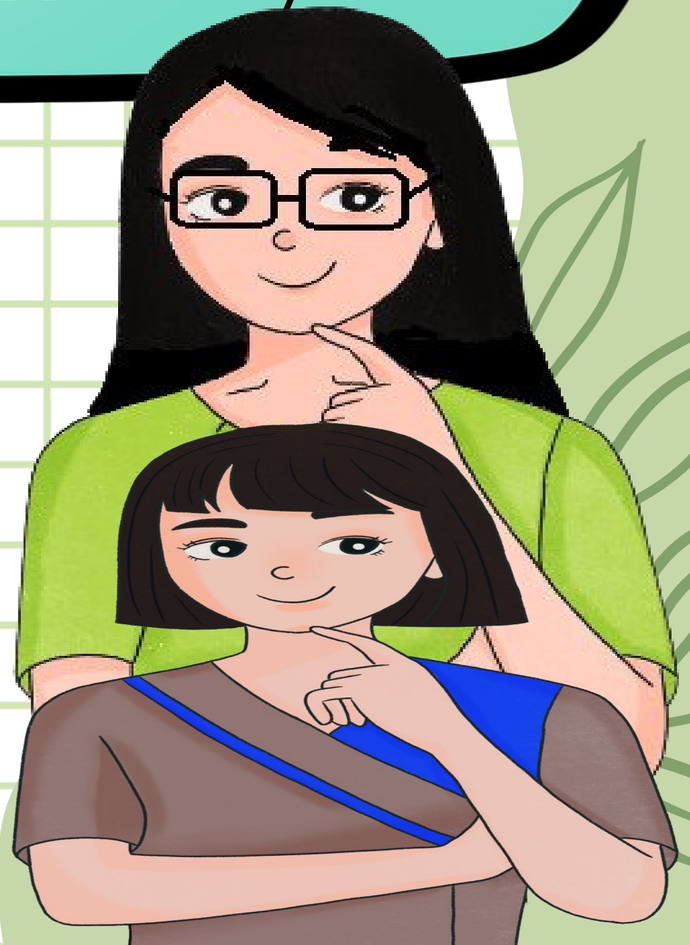
Read each statement carefully. Write True if the statement is correct and False if it is incorrect.

An electrical permit is required for installing any electrical portable equipment, regardless of its power rating.

TRUE

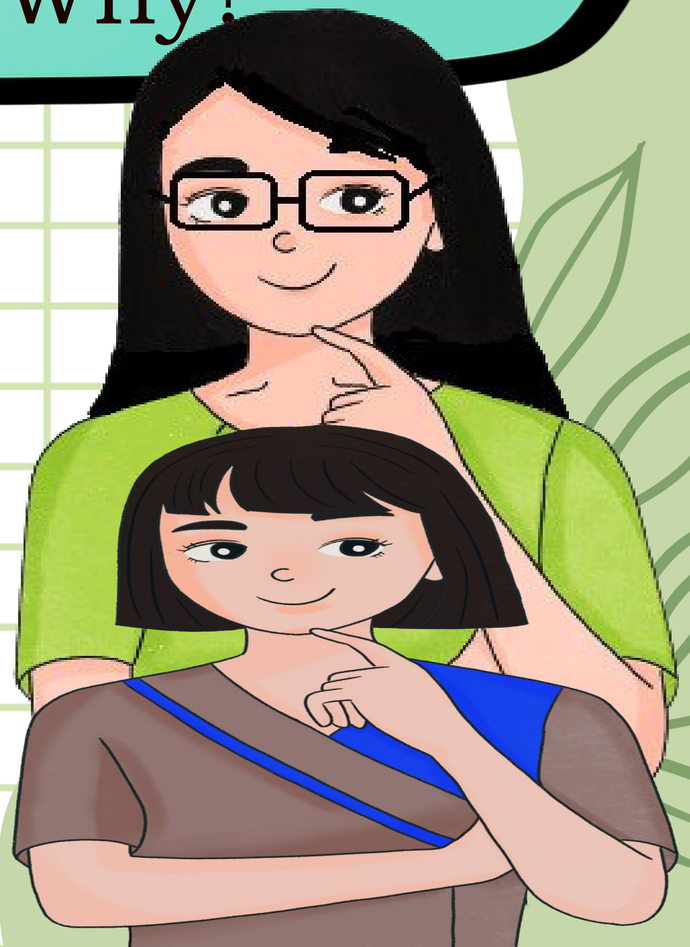
FEEDBACK

Do you have a community builder in your family?



FEEDBACK

If given the opportunity, which career path in industrial arts will you pursue? Why?



Lesson Purpose

What is the
cture all about?

house on fire

electric shock

traffic accident

Lesson Purpose

What do you think is our topic for today?

house on fire

electric shock

traffic accident

Lesson Purpose

- Faulty wiring or overloaded circuits can cause fires. The PEC sets safety standards to prevent this.
- Exposed wires or poor grounding can lead to electrocution. The PEC ensures proper installation for safety.
- Downed power lines or faulty street lighting can be electrical hazards. The PEC regulates safe installations.

Philippine Electrical Code (Residential Wiring)

Quarter 4 Week 2 – Day 2

Content Standards

The learners demonstrate an understanding of the concepts in industrial arts services.

Performance Standards

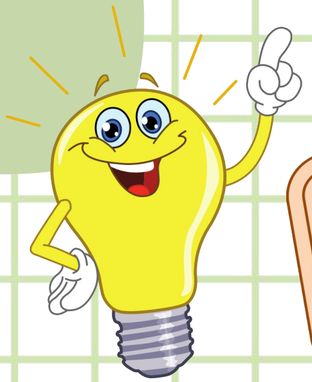
The learners perform mensuration and calculations following safety precautions.

Learning Competencies

- Discuss the basic concepts of the building code, electrical code, electronics code, traffic, and road signs.
- Explain the importance of following the set codes and standards for industrial arts services.

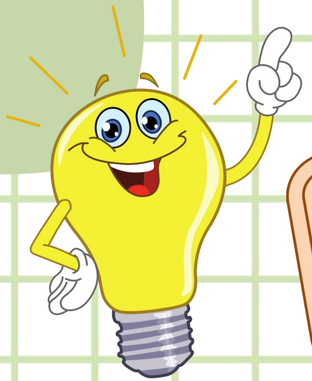
Learning Objectives

1. Describe the legal bases behind electrical code in the Philippine setting.
2. Enumerate basic requirements and standards for electrical code of the Philippines.
3. Recognize to the importance of following the set codes and standards for electrical codes in daily life.



PERMIT

an official document giving someone authorization to do something.



DWELLING

a house, apartment, or other place of residence.

Explicitation

- What is the Philippine Electrical Code (PEC)?
- What government agency/ies is responsible for implementing and regulating PEC?
- What are the basic requirements in residential wiring?

LET'S FIND OUT!

The **Philippine Electrical Code (PEC)** is a set of standards and regulations that govern the safe and proper installation, operation, and maintenance of electrical systems in the Philippines. It provides **guidelines** for electrical design, wiring, equipment selection, grounding, protection, and other related aspects to ensure electrical safety and efficiency.

Objectives

a) Safe Use of Electricity – to establish basic materials quality and electrical work standards for the safe use of electricity for light, heat, power, communications, signaling and for other purposes.

Objectives

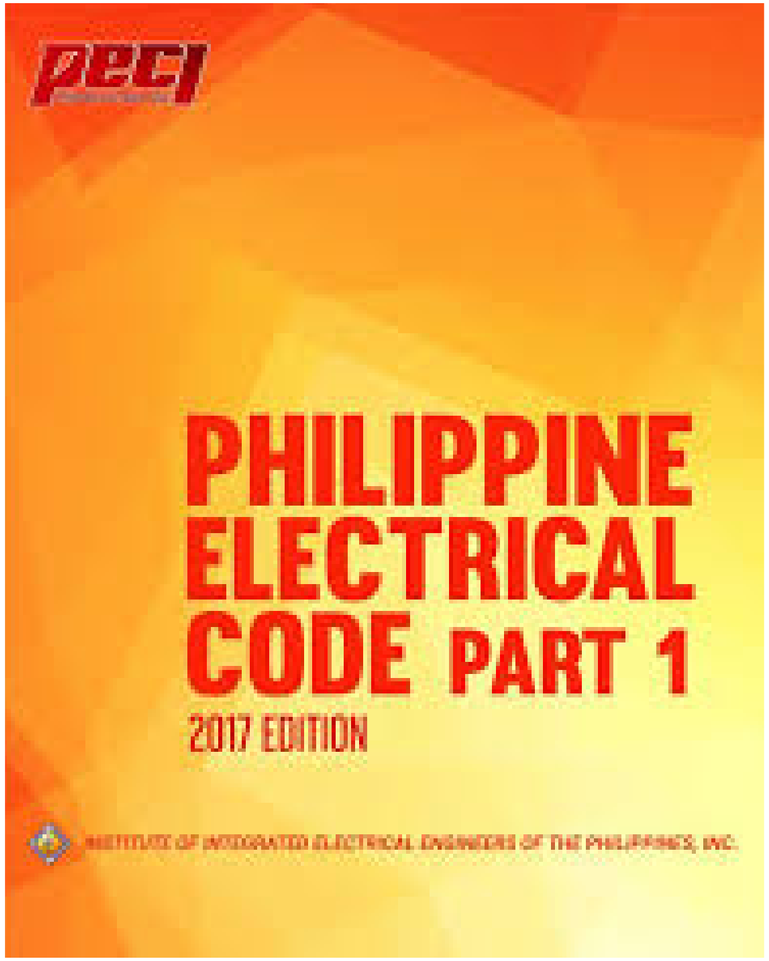
b) Adequacy – strict compliance with the provisions of this Code will ensure safety in electrical installation and construction, but not necessarily efficient, convenient, or adequate for good service or future expansion of electrical use.

The use of PEC is mandated by R.A. 7920 (formerly R.A. 184), also known as the New Electrical Engineering Law. Likewise, referral codes are also used which are mandated by their corresponding laws.

1. P.D. 1096 – National Building Code
2. P.D. 1185 – Fire Code of the Philippines
3. Structural Code R.A. 7920 – is an act providing for a more responsive and comprehensive regulation for the practice, licensing and registration of electrical engineers and electricians.

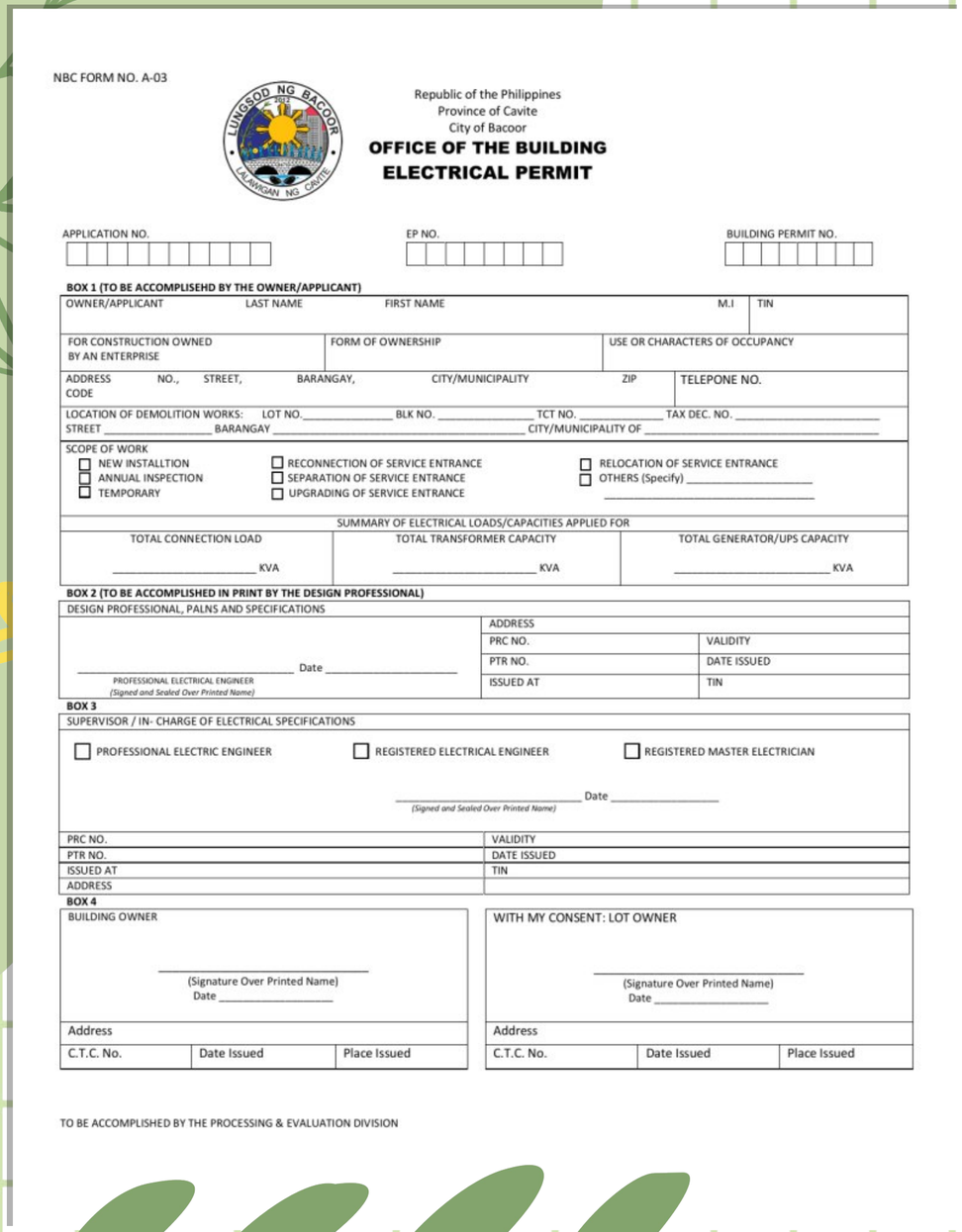
Government Authorities who implement PEC

1. Office of the Building Official
2. Office of the City Electrician (City Electrical Supervising Engineer)
3. Office of the Municipal Electrician (Municipal Electrical Supervising Engineer)
4. Regional Office of the DOLE



The local Building Officials are responsible in enforcing the provisions of the Philippine Electrical Code as indicated in RA 7920, Sec 39 and PD 1096, Sections 205 and 1301.

The role of the **Bureau of Fire Protection** (BFP) as per provisions of the Republic Act No. 9514(3), Fire Code regarding electrical systems is to enforce the provisions of the Fire Code particularly Section 7.A.1 to ensure that there are no violations as per Section 8.B. It has no legal authority on the design of electrical system.



NBC FORM NO. A-03

Republic of the Philippines
Province of Cavite
City of Bacoor

OFFICE OF THE BUILDING
ELECTRICAL PERMIT

APPLICATION NO. | EP NO. | BUILDING PERMIT NO.

BOX 1 (TO BE ACCOMPLISEHD BY THE OWNER/APPLICANT)

OWNER/APPLICANT | LAST NAME | FIRST NAME | M.I | TIN

FOR CONSTRUCTION OWNED BY AN ENTERPRISE | FORM OF OWNERSHIP | USE OR CHARACTERS OF OCCUPANCY

ADDRESS NO., STREET, BARANGAY, CITY/MUNICIPALITY ZIP CODE | TELEPONE NO.

LOCATION OF DEMOLITION WORKS: LOT NO._____ BLK NO. _____ TCT NO. _____ TAX DEC. NO. _____
STREET _____ BARANGAY _____ CITY/MUNICIPALITY OF _____

SCOPE OF WORK
- ☐ NEW INSTALLTION
- ☐ ANNUAL INSPECTION
- ☐ TEMPORARY
- ☐ RECONNECTION OF SERVICE ENTRANCE
- ☐ SEPARATION OF SERVICE ENTRANCE
- ☐ UPGRADING OF SERVICE ENTRANCE
- ☐ RELOCATION OF SERVICE ENTRANCE
- ☐ OTHERS (Specify) _____

SUMMARY OF ELECTRICAL LOADS/CAPACITIES APPLIED FOR

TOTAL CONNECTION LOAD	TOTAL TRANSFORMER CAPACITY	TOTAL GENERATOR/UPS CAPACITY
_____ KVA	_____ KVA	_____ KVA

BOX 2 (TO BE ACCOMPLISHED IN PRINT BY THE DESIGN PROFESSIONAL)

DESIGN PROFESSIONAL, PALNS AND SPECIFICATIONS

_____ Date _____
PROFESSIONAL ELECTRICAL ENGINEER
(Signed and Sealed Over Printed Name)

ADDRESS	
PRC NO.	VALIDITY
PTR NO.	DATE ISSUED
ISSUED AT	TIN

BOX 3

SUPERVISOR / IN- CHARGE OF ELECTRICAL SPECIFICATIONS

☐ PROFESSIONAL ELECTRIC ENGINEER ☐ REGISTERED ELECTRICAL ENGINEER ☐ REGISTERED MASTER ELECTRICIAN

_____ Date _____
(Signed and Sealed Over Printed Name)

PRC NO.	VALIDITY
PTR NO.	DATE ISSUED
ISSUED AT	TIN
ADDRESS	

BOX 4

BUILDING OWNER

_____
(Signature Over Printed Name)
Date _____

Address

C.T.C. No.	Date Issued	Place Issued

WITH MY CONSENT: LOT OWNER

_____
(Signature Over Printed Name)
Date _____

Address

C.T.C. No.	Date Issued	Place Issued

TO BE ACCOMPLISHED BY THE PROCESSING & EVALUATION DIVISION

Permits and Inspection Certificates

A permit is required before undertaking any electrical installation. An inspection is also required after which certificate of final electrical inspection (CFEI) is issued by the authority.

Exceptions

A permit is not required for;

1. the installation of electrical portable equipment rated not more than 1,200 VA.
2. reconnection of disconnected service due to non-payment of electric bill or change of occupants for a period of one year.

EXPLICITATION

1. What is the Philippine Electrical Code (PEC)?

RA 7920 – New Electrical Engineering Law, P.D. 1096 – National Building Code, P.D. 1185 – Fire Code of the Philippines and R.A. 7920 – Structural Code of the Philippines

EXPLICITATION

2. What government agency/ies is responsible for implementing and regulating PEC?

A. Office of the Building Official
B. Office of the City Electrician (City Electrical Supervising Engineer)
C. Office of the Municipal Electrician (Municipal Electrical Supervising Engineer)
D. Regional Office of the DOLE E. Bureau of Fire Protection to enforce the provisions in the fire code.

EXPLICITATION

3. What are the basic requirements in residential wiring?

A permit is required before undertaking any electrical installation. An inspection is also required after which certificate of final electrical inspection (CFEI) is issued by the authority.

LESSON ACTIVITY:

T-I-L-T

Things I Learned Today, with a Pair!
Directions: In the boxes provided, the students will answer the following essential questions:

LESSON ACTIVITY:

What?

Who?

How?

LESSON ACTIVITY:

1. What is the legal bases of the Philippine Electrical Code?
2. Who are the government agencies implementing PEC?
3. How is our lesson today connected to the previous lesson?

ANSWER

1. Legal Basis of the Philippine Electrical Code – The Philippine Electrical Code (PEC) is based on Republic Act No. 7920, also known as the New Electrical Engineering Law, which governs electrical installations and safety standards in the Philippines.

ANSWER

2. Implementing Agencies of PEC

A. Office of the Building Official

B. Office of the City Electrician (City Electrical Supervising Engineer)

C. Office of the Municipal Electrician (Municipal Electrical Supervising Engineer)

D. Regional Office of the DOLE E. Bureau of Fire Protection to enforce the provisions in the fire code.

ANSWER

3. This lesson builds on the National Building Code, focusing on residential wiring regulations under the PEC to ensure safe and efficient electrical installations in homes.

3-2-1 Game

3 Things I learned.

_____.

2 New words I encounter

_____.

1 Question I have in mind

_____.

Reflection on Learning

The students will accomplish the weekly reflection log.

1. My most favorite activity this day was:_____
2. This day I am proud of: _____
3. Next lesson, I want to improve on: _____

Formative Assessment

Multiple-Choice

Instructions: Choose the correct answer to each question.

Formative Assessment

1. The Philippine Electrical Code (PEC) covers electrical protection measures against:
a. effects of lightning.
b. accidental contact with power lines.
c. voltages/ electromagnetically /electrostatically induced into communication circuits by normal or fault currents in parallel runs of power lines.
d. all of the above

Formative Assessment

2. Which law mandates the use of the Philippine Electrical Code (PEC)?

a. P.D. 1185 – Fire Code of the Philippines

b. R.A. 7920 – New Electrical Engineering Law

c. P.D. 1096 – National Building Code

d. R.A. 9514 – Fire Code of the Philippines

Formative Assessment

3. Who is responsible for enforcing the provisions of the PEC at the local level?

a. Department of Energy (DOE)

b. Bureau of Fire Protection (BFP)

c. Office of the Building Official

d. Professional Regulation Commission (PRC)

Formative Assessment

4. When is an electrical permit NOT required?
a. For new residential electrical installations
b. For electrical portable equipment rated not more than 1,200 VA
c. For upgrading an old electrical system
d. For commercial electrical wiring installation

Formative Assessment

5. What is the role of the Bureau of Fire Protection (BFP) in relation to electrical systems?

a. To design electrical systems for public buildings

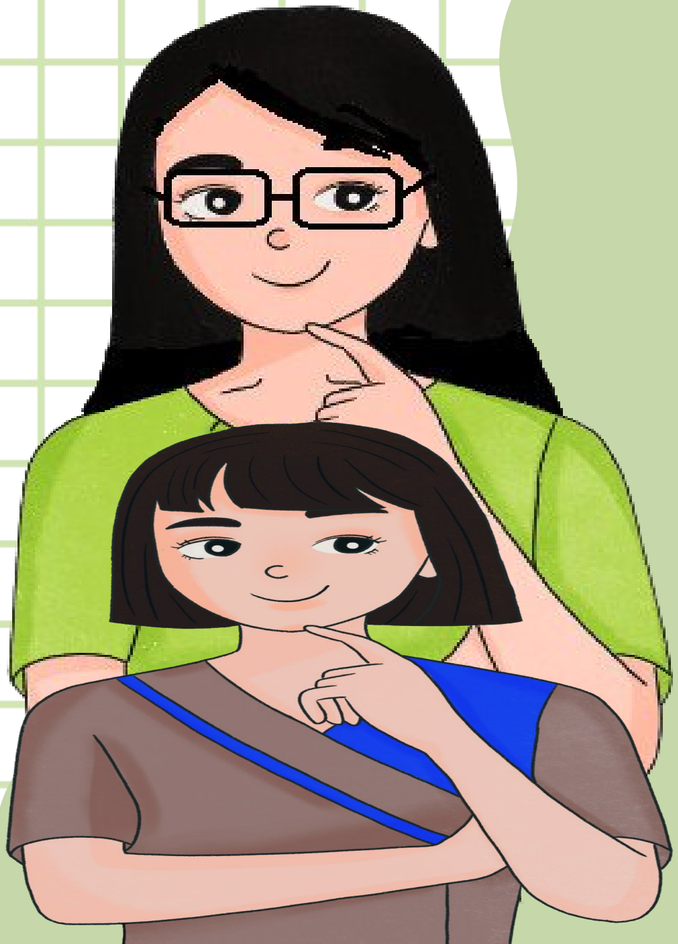
b. To approve electrical permits for residential buildings

c. To ensure compliance with fire safety regulations related to electrical systems

d. To conduct electrical engineering board examinations

Key to Correction

1. D
2. B
3. C
4. B
5. C



HOMEWORK

Continue to read and study the Electrical Code of the Philippines.

Thank You